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Drilling Performance Optimization in Unconventional Oil and Gas Wells in the UAE: Advanced Well Design and Drilling Technologies

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Abstract

The UAE is expanding its unconventional oil and gas drilling activities to unlock resources in tight gas, shale, and complex carbonate formations. However, these reservoirs demand innovative well designs and advanced drilling technologies to overcome challenges such as high uncertainties in formation pore pressure and a narrow window between fracture and pore pressures leading to elevated risks of wellbore instability, kicks, and mud losses, hard abrasive formations, complex geology, high-pressure/high-temperature (HPHT) conditions, and wellbore instability. This study focuses on optimizing drilling performance through integrated well planning, advanced drilling technologies and real-time operational adjustments tailored to the UAE's unconventional plays.

The research evaluates fit-for-purpose adaptive well designs (e.g., optimized casing programs and trajectory planning), including extended-reach horizontal wells, combined with high-performance drill bits, Drillstring vibration mitigation tools, hybrid mud systems, and automated drilling control. Advanced technologies such as managed pressure drilling (MPD), and high-frequency downhole sensors are analyzed to enhance drilling efficiency and reduce non-productive time (NPT). Field data from UAE's unconventional basins are analyzed to identify inefficiencies and optimize drilling parameters used to benchmark key performance indicators (KPIs) like rate of penetration (ROP), torque/drag reduction, and hole quality.

Field implementation across the UAE unconventional wells demonstrated transformative improvements:

1. Rate of Penetration (ROP) Enhancement: Achieved 250% increase in abrasive zones through: High-efficiency PDC bits with Real-time ML-based parameter optimization and Downhole vibration dampeners reducing lateral shocks by 70% and minimize drillstring failures.
2. Well Delivery Time Reduction: Slashed total drilling time by 67% reduction in some concessions. Radical Time Compression is Achievable proving that process redesign and automation can overcome geological complexity without sacrificing safety.
3. Cost Optimization with 30% lower well costs) attributed to Reducing rig time and minimizing casing strings through Geomechanical modeling reducing contingency strings and Zero wellbore instability NPT. All this cost reduction will minimize the total cost per barrel.

The integration of tailored well designs, cutting-edge drilling technologies, and digital solutions significantly enhances drilling performance in UAE's unconventional reservoirs. This framework provides actionable insights for operators to achieve cost-effective and sustainable development.

Introduction

The United Arab Emirates (UAE), traditionally recognized for its prolific conventional oil reserves, has recently intensified its efforts to develop unconventional oil and gas resources. The UAE is actively developing unconventional hydrocarbon resources to enhance energy security and meet rising demand. In 2020, it was announced the first production of shale gas (from the Ruwais Diyab unconventional concession) and set a target of 1 Bcf/d of domestic gas by 2030. Similarly, Abu Dhabi's Late Cretaceous carbonate source rock Formation is being appraised for shale oil potential. These reservoirs are characterized by ultra-tight pore space, high pressure, and complex geomechanics, which pose drilling challenges (wellbore instability, stuck pipe, lost circulation). Consequently, optimizing drilling performance is imperative to make these plays economic. This requires integrating advanced pre-well planning (trajectories guided by seismic and Geomechanical models), custom BHA and fluid design, and cutting-edge drilling technology. Recent efforts have deployed RSS, premium bits, and real-time data analytics to set new benchmarks: a well was drilled in less than 14 days (67 % reduction in some concessions) and the pad campaign achieved 53% reduction in average well time. Radical Time Compression is Achievable proving that process redesign and automation can overcome geological complexity without sacrificing safety. Cost Optimization with 30% lower well costs attributed to Reducing rig time and minimizing casing strings through Geomechanical modeling reducing contingency strings and Zero wellbore instability NPT. All this cost reduction will minimize the total cost per barrel. Additionally, the application of AI and digital platforms is growing that AI tools (for drilling, reservoir, and equipment monitoring) generated \$500 M in value in 2023. This paper provides a comprehensive overview of drilling performance optimization approaches currently deployed or under evaluation in the UAE's unconventional fields (Jacobs, 2020).

Field Background

The stratigraphic column of the UAE is shown in [Figure 1](#), unconventional oil reservoirs are present in Late Cretaceous carbonate source rock formation while unconventional gas reservoirs in Diyab formations ([Ali et al., 2013](#); [Van Laer et al., 2020](#)).

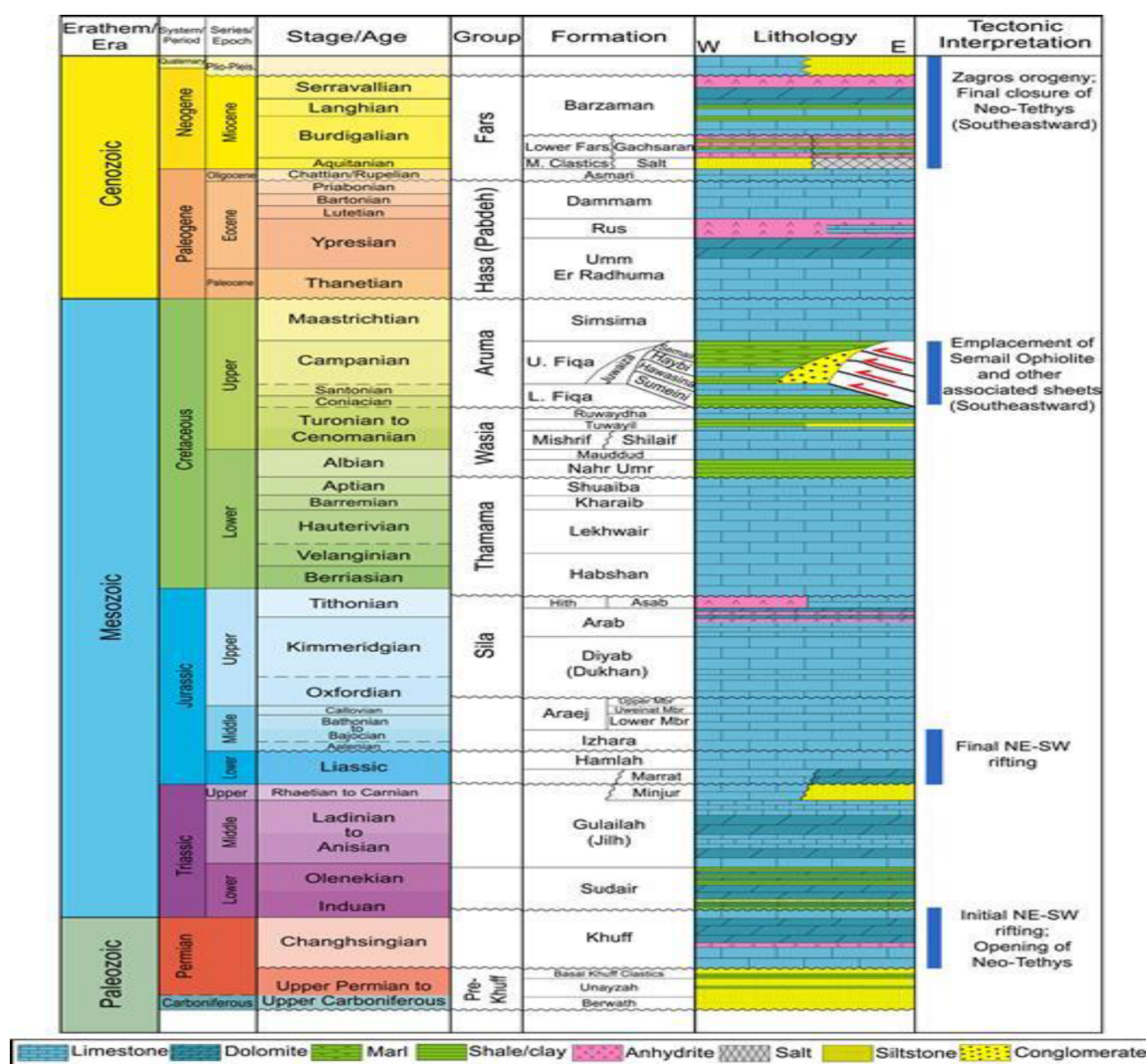


Figure 1—UAE Stratigraphic column (modified after Ali et al. (2013), Alsharhan (1989), and Hu et al. (2016))

Technical Challenges in UAE Unconventional Reservoirs

Drilling unconventional wells in the UAE poses unique technical hurdles.

- **Narrow Pressure Margins:** A slim window between pore and fractured pressure increases the risk of kicks and mud losses.
- **Wellbore Instability:** Poor hole cleaning and inadequate Geomechanical modeling contribute to borehole collapse, pack-off, or stuck pipe.
- **HPHT Conditions:** Elevated downhole temperatures ($>150^{\circ}\text{C}$) and pressures ($>10,000$ psi) impact tool reliability and drilling fluid stability.
- **Complex Lithology:** Interbedded tight carbonates, chert, and shale layers with high abrasiveness increase drillstring wear and reduce ROP.
- **High compressive strength:** reached more than 25k ksi which requires special bit design to achieve higher ROP and minimize premature bit failure.
- **Well control issues and gas flow:** high percentage of gases during drilling reservoir section in gas wells.
- **High H₂S content:** during drilling reservoir formations which require sour service tools and special mud products.

- Uncertainties in seismic: high pressure zone, unconsolidated and naturally fractured formations due to faults

Late Cretaceous carbonate source rock Formation (Unconventional Oil)

The Late Cretaceous carbonate source rock Formation (Cenomanian) is an organic-rich carbonate source rock underlying parts of UAE. It is the primary source for oil reserves and is now targeted directly as a tight-oil reservoir. Regional studies show Late Cretaceous carbonate source rock deposited in a deeper water intrashelf basin. Thickness of non-eroded Late Cretaceous carbonate source rock ranges ~500–900 ft in Abu Dhabi's deep basin, thinning to ~100–400 ft toward shelf edges. The formation comprises 3–4 stacked depositional sequences, each containing calcispheres and foraminiferal calcite with up to ~10% TOC. Porosity is dominated by nanopores (max ~15%) and permeability is extremely low (nano- to microdarcies). Some Late Cretaceous carbonate source rock sections remain 'wet' (above oil-water contacts), and test wells have produced "dry oil" from the shale itself. Because of ultra-low permeability, Late Cretaceous carbonate source rock production requires long horizontals with extensive hydraulic fracturing. These results have validated the resource potential of Late Cretaceous carbonate source rock and demonstrated the need for coordinated geological, drilling, and completion workflows in its development (Ali et al., 2013; Van Laer et al., 2020).

Upper Jurassic Gas Plays (Diyab)

The Upper Jurassic (Oxfordian–Kimmeridgian) carbonate source rock throughout Abu Dhabi (Ruiz et al., 2019; Al-Hajari & Al-Marzouqi, 2023). It is naturally overpressured and highly carbonate-rich (>80% calcite, <15% clay), making it exceptionally brittle and amenable to fracturing but challenging to drill. This interval can be subdivided into three main reservoir units. Each contains organic-rich carbonate mudstones (TOC often 2–3%) interbedded with tight limestones. These units served as source rocks for the giant Ghawar/Umm Shaif fields and are now targeted as unconventional gas reservoirs. Thick overpressure (~0.65 VRr in Abu Dhabi's synclines) and regional NE–SW yield a strike-slip stress regime; minimum horizontal stress is close to overburden in many areas. Geomechanical studies indicate vertical stress ~1.08 psi/ft, with minimum horizontal ~0.95–1.08 psi/ft and maximum horizontal ~1.10–1.17 psi/ft, meaning many potential stress barriers (e.g. carbonate benches) exist within the section. In aggregate, the Jurassic carbonates of the Diyab play present enormous gas potential but demand careful drilling design and stimulation to be produced economically.

Methodology: Drilling Optimization Workflow

To optimize drilling in these tight carbonate plays, a systematic workflow is followed for each well section (surface to target and horizontal). Key elements are:

- **Well Design:** The casing setting depth is very critical per each section to minimize the drilling problems and challenges. Incorporation of high-fidelity modeling reduces unnecessary casing strings, ensuring zonal isolation while minimizing material costs. Changing the casing design from heavy casing design (HCD) to light casing design (LCD) leading to saving the tangible costs by 30% and also leads to minimize the well duration by more than 35% which reflects on well cost saving reached 30% [Figure 2](#) & [Figure 3](#)
- **Trajectory Planning:** Extensive pre-drill planning uses 3D seismic, structural maps, and Geomechanical models to define optimal paths. Landing zones are selected in the thickest organic-rich intervals (identified by logs and seismic facies). Well profiles often include long horizontal legs (6000 ft MD and the new target is 10,000 ft MD) oriented azimuthally to avoid major faults and align with the minimum stress azimuth. Build/drop rates are modeled to minimize tortuosity while achieving reservoir contact. Geosteering capability (real-time gamma/porosity logs) is prepared to

keep the bit in the sweet spot. Literature notes that directional planning often involves balancing slide drilling (to navigate targets) and rotary drilling (for efficiency and hole quality). The trajectory of the well playing also main role in the final friction factors by minimizing the well tortuosity and side forces which leads to minimizing the friction factor while drilling and while run in hole with frac string for this long horizontal section. Negative displacement was applied to maximize the reservoir exposure and help with the run-in hole with the frac string. [Figure 5](#) shows example for trajectory of pad consisting of 4 wells each well targeting different formation or different sublayer in the formation.

- **BHA Design:** The bottom-hole assembly is tailored for the formation. High-torque premium bits (e.g. PDC or Hybrid designs) are chosen for hard limestone/calcite. Downhole motors or rotary steerable systems (RSS) are used to drill horizontals while maintaining constant rotation, The use of RSS contributed to achieving record drilling times (well delivery in <14 days) and smoother horizontal drilling without the frequent slide/rotate cycles needed by conventional motors. Stabilizer placement and reamer strategy ensure gauge hole. Shocks and vibrations are also one of the key challenges for BHA design as it affects directional tool failures, controlling drilling parameters which will reflect on the rate of penetration (ROP) that is the reason a lot of shock dampening tools have been tried and evaluated to select the optimum tool. Placement of downhole shock and torsional dampeners [Figure 6](#) & [Figure 7](#) reduced lateral shocks by 70%, minimizing drillstring failures and apply more drilling parameters which lead to improve ROP by 30% [Figure 8](#) . Measurement-While-Drilling (MWD) and Logging-While-Drilling (LWD) tools provide downhole information (inclination, azimuth, gamma, resistivity, LWD porosity, etc.) in real time. This allows geosteering within the target zone and immediate detection of lithology changes. Integration of MWD/LWD data with surface rig telemetry enables real-time analytics of drilling parameters (e.g. ROP, torque, vibration). Advanced rigs now use automated tracking of drilling data to identify inefficiencies. In practice, drilling teams use centralized dashboards to monitor downhole sensors and adjust parameters (WOB, RPM, mud pump rates) on the fly, improving efficiency and preventing incidents. After early wells, teams review bit wear and dynamics to refine BHA for each section. Standard BHA has been selected for each section moreover road map for each formation has been established to maximize the rate of penetration, minimize bit wear and premature failure, bit trips and drillstring tool failure.
- **Drilling Fluids:** Custom drilling fluids are used to maintain wellbore stability and optimize hydraulics. A viscous, low-solid fluid may be used to suspend cuttings and minimize differential sticking in low-porosity rock. Mud weight is set via geomechanical analysis to straddle the narrow window between pore pressure and fracture pressure. Lubricants and friction reducers help lower torque in long horizontals. Mud type and mud chemical selection is a critical process due to the high temperature and complex formation heterogeneity. OBM is being used in reservoir section in some fields to mitigate shale problems. Meanwhile HPWBM is being used in other fields due to the risk of losses and the presences of high percentages of gases in HTHP conditions. Bridging materials and Micronized barite are being used to overcome differential sticking and filter cake thickness meanwhile strengthening weak unconsolidated formations. The application of mud coolers and chillers to minimize the mud temperature to mitigate directional drilling failures ([Schlumberger, 2023](#)).
- **Geomechanical Modeling:** A detailed mechanical earth model (MEM) is built from log and core data from offset wells [Figure 9](#) to predict stresses, brittleness, and fracture barriers [Figure 10](#). This guides decisions on safe mud weights (to avoid lost circulation or fracturing) and landing targets away from highly stressed zones. For example, models often identify a hard carbonate "bench" between reservoir sublayers that could act as a stress barrier to fracture propagation. The MEM also

informs fracturing strategies post-drilling. Overall, geomechanics is key to preventing wellbore instability and minimizing NPT (Al-Hajari & Al-Marzouqi, 2023; Lazreq et al., 2020).

- Real time drilling data: real time drilling data Figure 11 from the wells allows detecting drilling anomalies at the same time which helps for better decision making to solve drilling problems like Drilling string Stuck, formation losses, well control issues, etc. Also it helps for checking the application of the correct drilling parameters per section to maximize the ROP and minimize drillstring and bit failures.
- KPI and benchmarking the rig performance: Invisible lost time and flat time in drilling operations represents more than 30% from the total well duration. By benchmarking each operation in the drilling process we can track the invisible lost time and minimize or eliminate the ILT. Example Weight to weight connection time has been decreased from 30.7 min to less than 4.9 min as shown in Figure 12
- New technology and the application of AI: Artificial intelligence and machine learning algorithms are increasingly applied to optimize drilling. Data-driven models can predict optimal drilling parameters (bit weight, RPM) to maximize ROP and minimize undesirable vibration. In UAE, the operating company reports deploying over 30 AI-driven tools across operations; in upstream these include applications for mapping subsurface, optimizing drilling trajectories, and managing drilling equipment. For example, AI can analyze historical rig data to forecast drilling hazards (e.g. torque spikes) and suggest preventative actions. AI enables Exploring algorithms for real-time ROP prediction and drill string vibration control to keep drilling on target efficiently by selecting the optimum parameters for each formation, there is a screen with dashboard connected to the driller cabin showing the driller the optimum parameters (WOB, RPM, differential pressure) for the formation also it is connected to the real time system and give alerts to the drilling team if the driller not following this road map and also showing the percentages of adherence to this road map. Downhole High-Frequency Sensors: Provided real-time vibration, pressure, and temperature data for proactive parameter adjustments (ADNOC, 2024; Zhang & Chen, 2021)
- Managed pressure drilling: Enabled precise pressure control in HPHT wells, mitigating influx and loss events. The implementation and introduction of MPD in the operating company's unconventional gas wells. Gas wells inherently pose unique challenges, and this complexity is amplified in unconventional reservoirs. These formations are often characterized by heterogeneous permeability and porosity, the presence of natural fractures and faults, and unpredictable wellbore stability. For the operating company, the initial foray into drilling such wells presented a steep learning curve, particularly in understanding and predicting reservoir behavior. Reservoir Pore pressure estimations received were high which required high mud weights as a precautionary measure to stay overbalance with less information on expected reservoir behavior. UC Gas PADs were drilled with no MPD there were numerous challenges observed such as tool failures due high mud weights, frequent operational slowdowns triggered by gas shows, which were treated with conventional well control responses, frequent stoppage to drilling operations and increase of mud weight due to downhole gasses, anomalies and gasses during trips, well securing and increasing mud weight, gas cut mud & circulating pressure fluctuations post check trips, ended with circulating through choke and stopping drilling, elevated circulating temperatures, high cost of mud & cement slurry due to required higher densities for displacement hierarchy and frack string cement displacement brine. MPD Figure 15 & Figure 16 was introduced and Pads were drilled with MPD with lots of benefits due to conducting Dynamic and static Pore pressure tests, formation characterization to find out operational boundaries, manage drilled, background, connection gasses while drilling or tripping without stopping or shut in procedures which led to drilling laterals with lower Mud weights, lower density cement slurries and displacement fluid to displace frack string cement. Exploration wells with pilot hole and further horizontal benefited from utilizing MPD

in pilot hole and later all apply knowledge in lateral and improve cost saving and operational complexities (Smith & Patel, 2022).

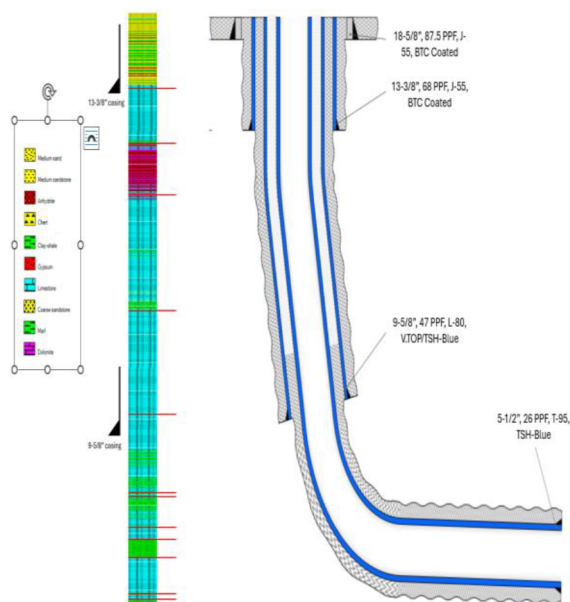


Figure 2—Oil Wells light casing design

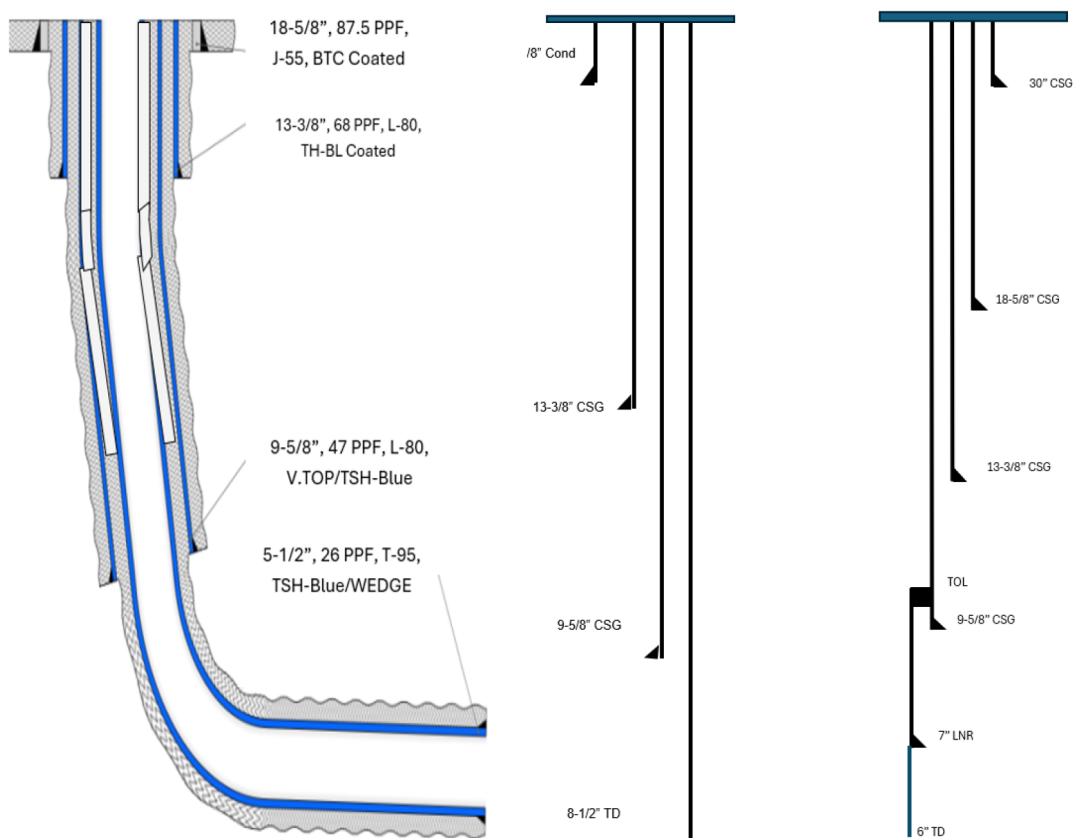


Figure 3—Comparison between LCD & HCD for gas wells

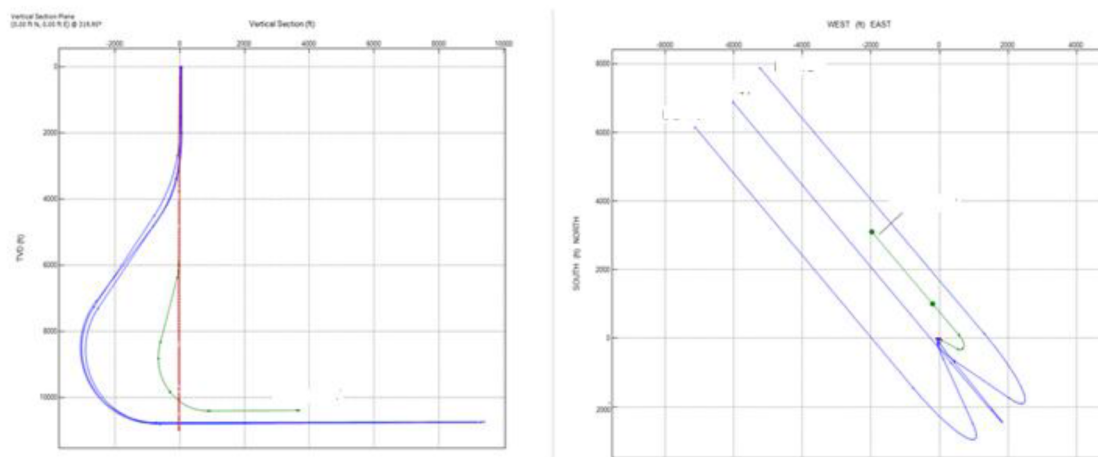


Figure 5—well trajectory for Pad Wells

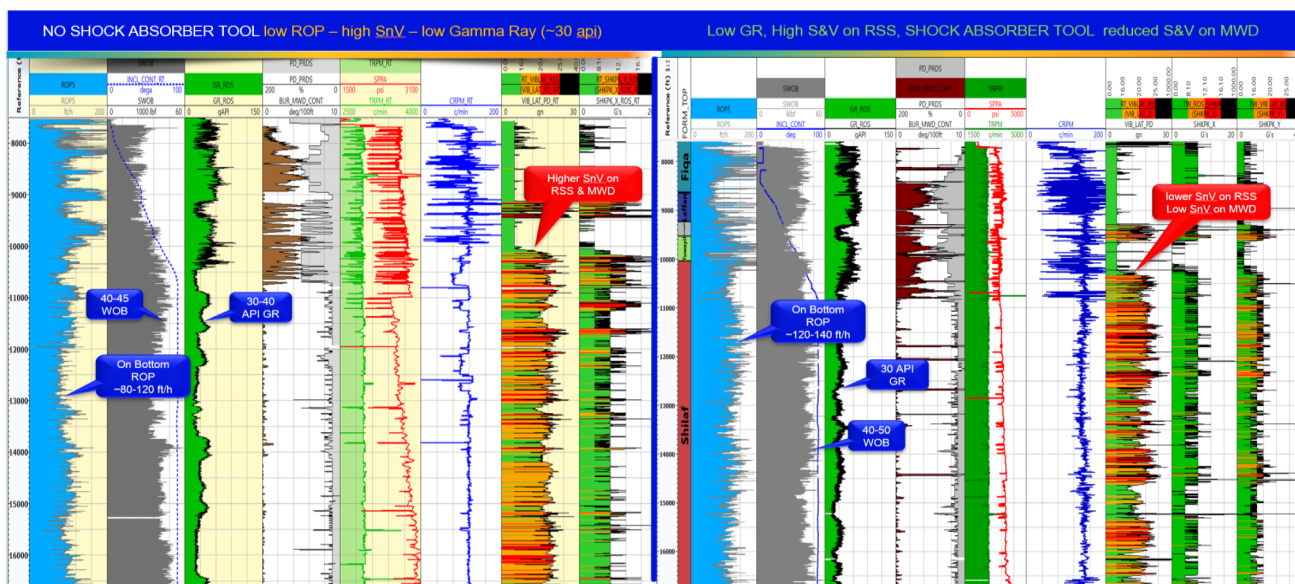


Figure 6—the effect of shock and torsional dampener tools

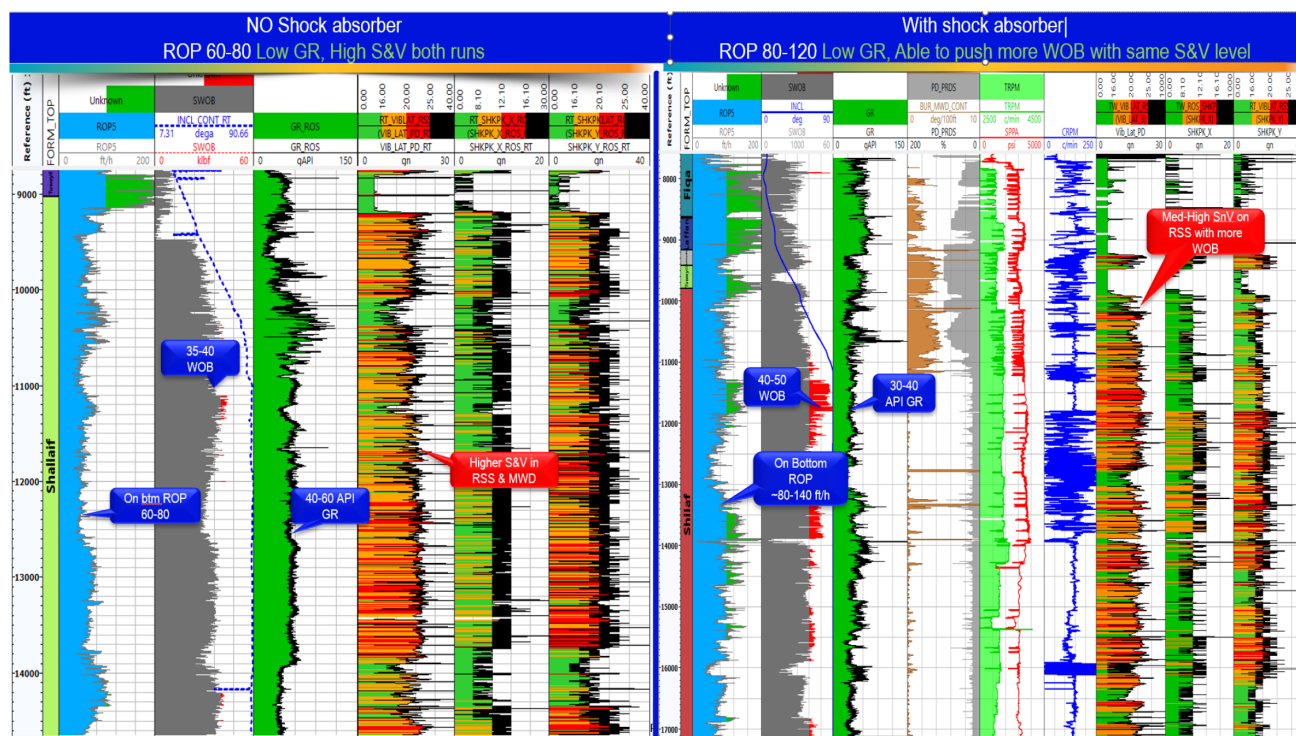


Figure 7—the effect of shock and torsional dampeners

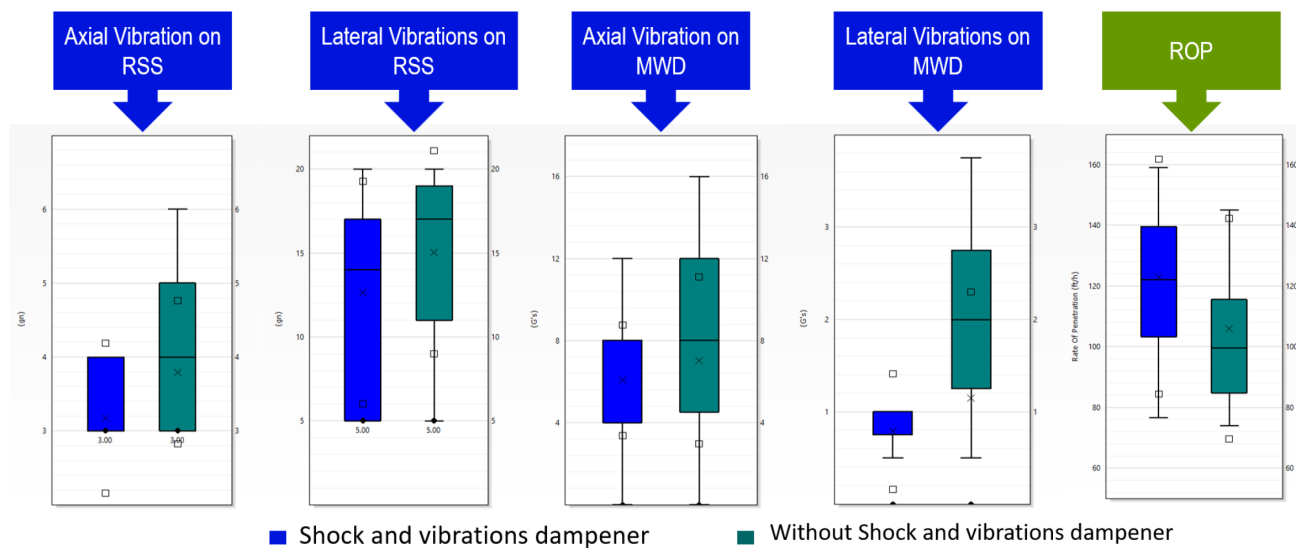


Figure 8—Comparison between shocks, vibrations and ROP with and without shock and vibrations dampener tools

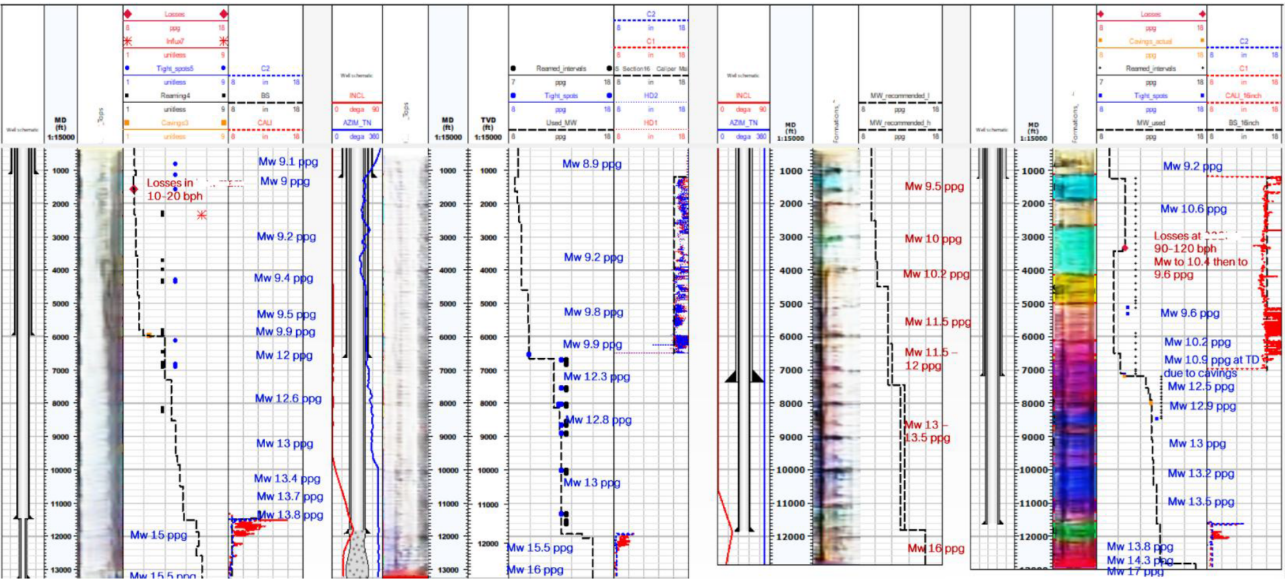


Figure 9—Geomechanical modelling from offset wells (Al-Hajari & Al-Marzouqi, 2023; Lazreq et al., 2020).

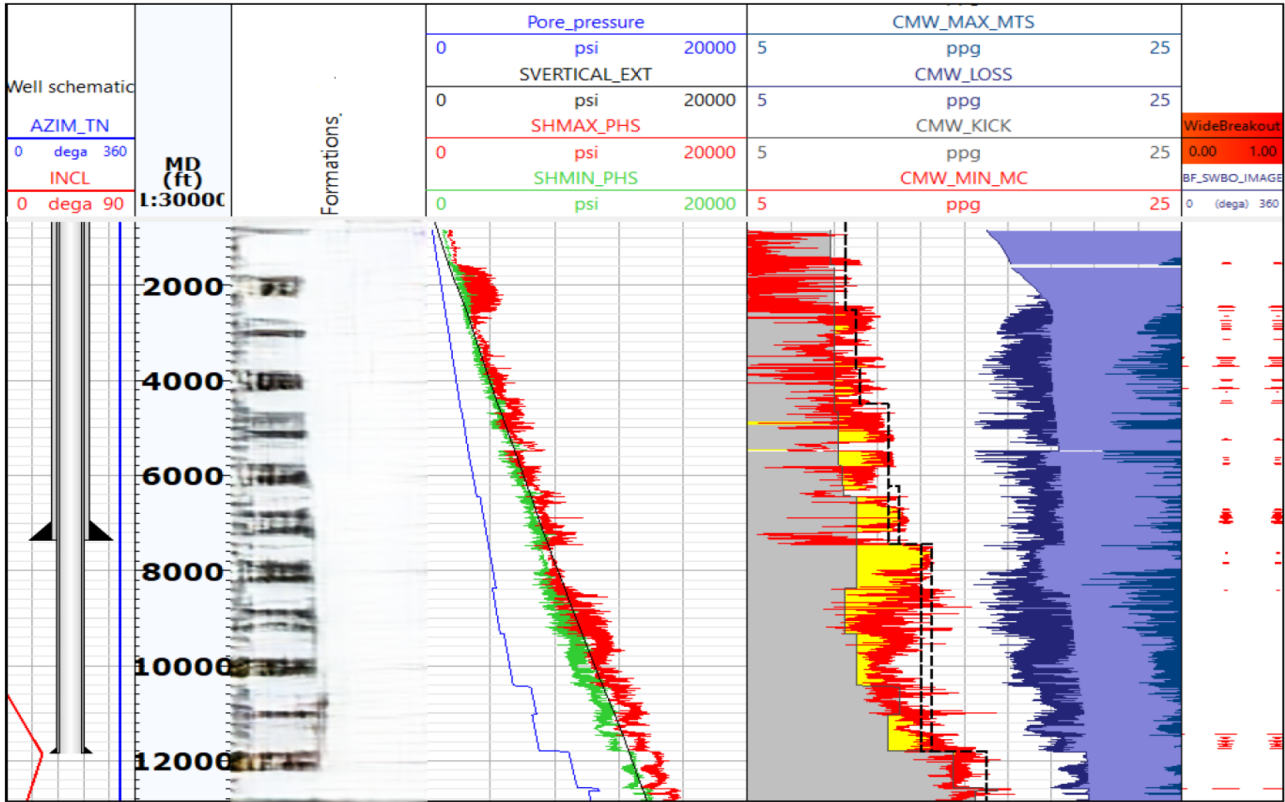


Figure 10—Predrill analysis using the predicted Geomechanical model (Al-Hajari & Al-Marzouqi, 2023; Lazreq et al., 2020)

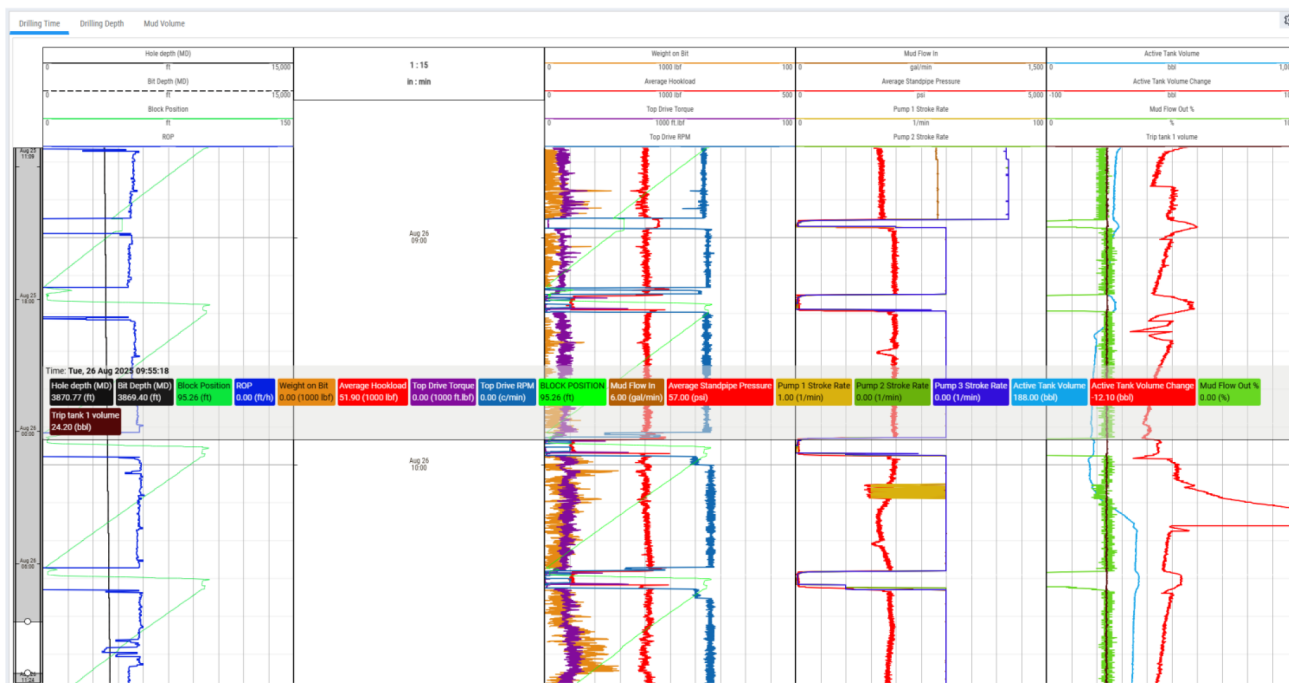


Figure 11—Real time data

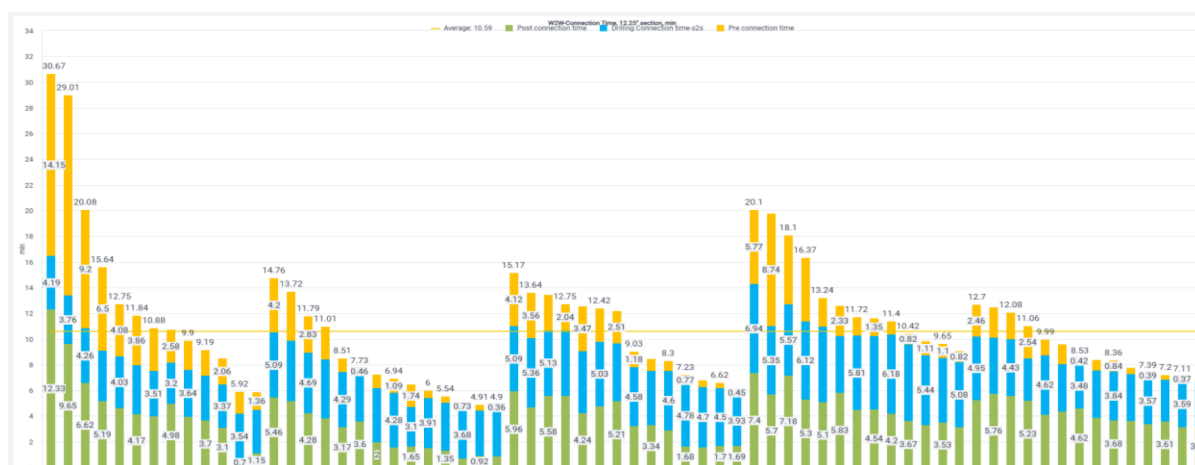


Figure 12—Oil Wells W2W Connection time for intermediate section

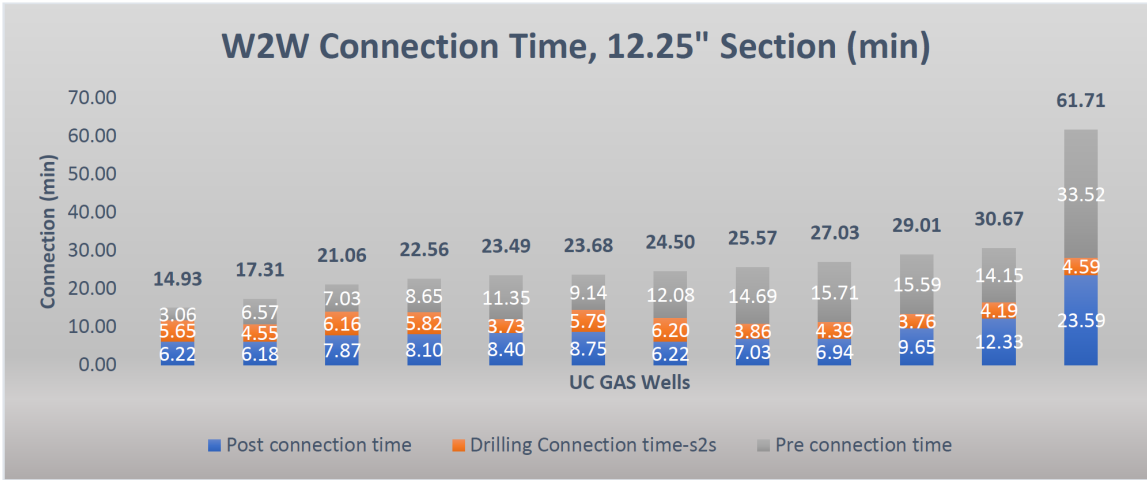


Figure 13—Gas wells W2W Connection time for intermediate section

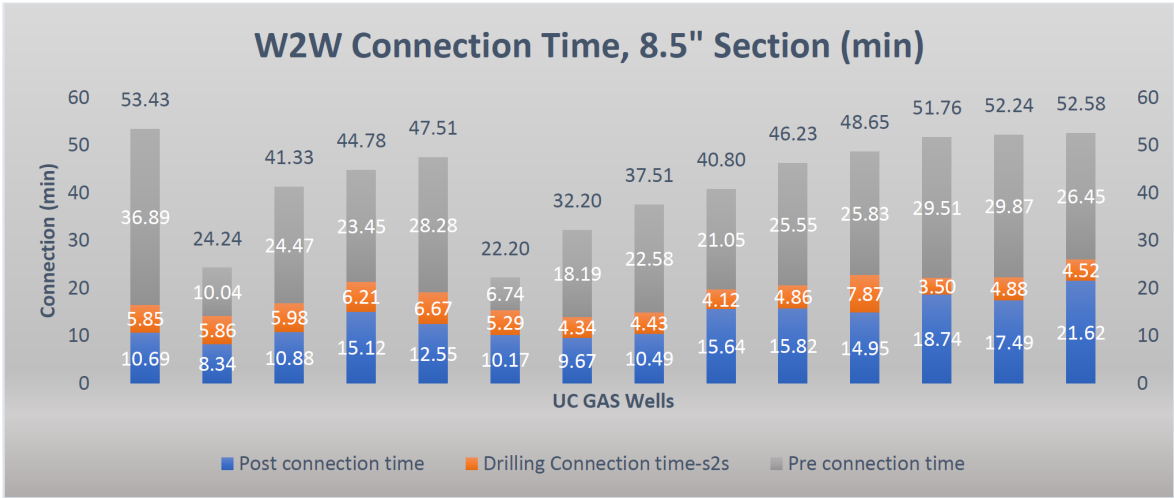


Figure 14—Gas wells W2W Connection time for Production section with and without MPD (Smith & Patel, 2022)

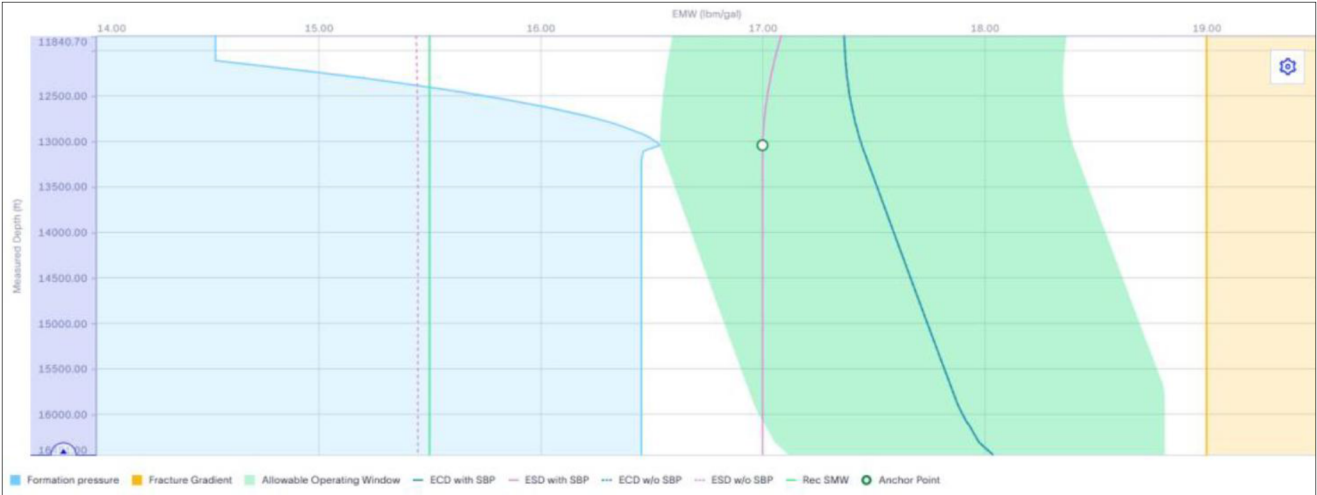


Figure 15—MPD Operating Window (Smith & Patel, 2022)

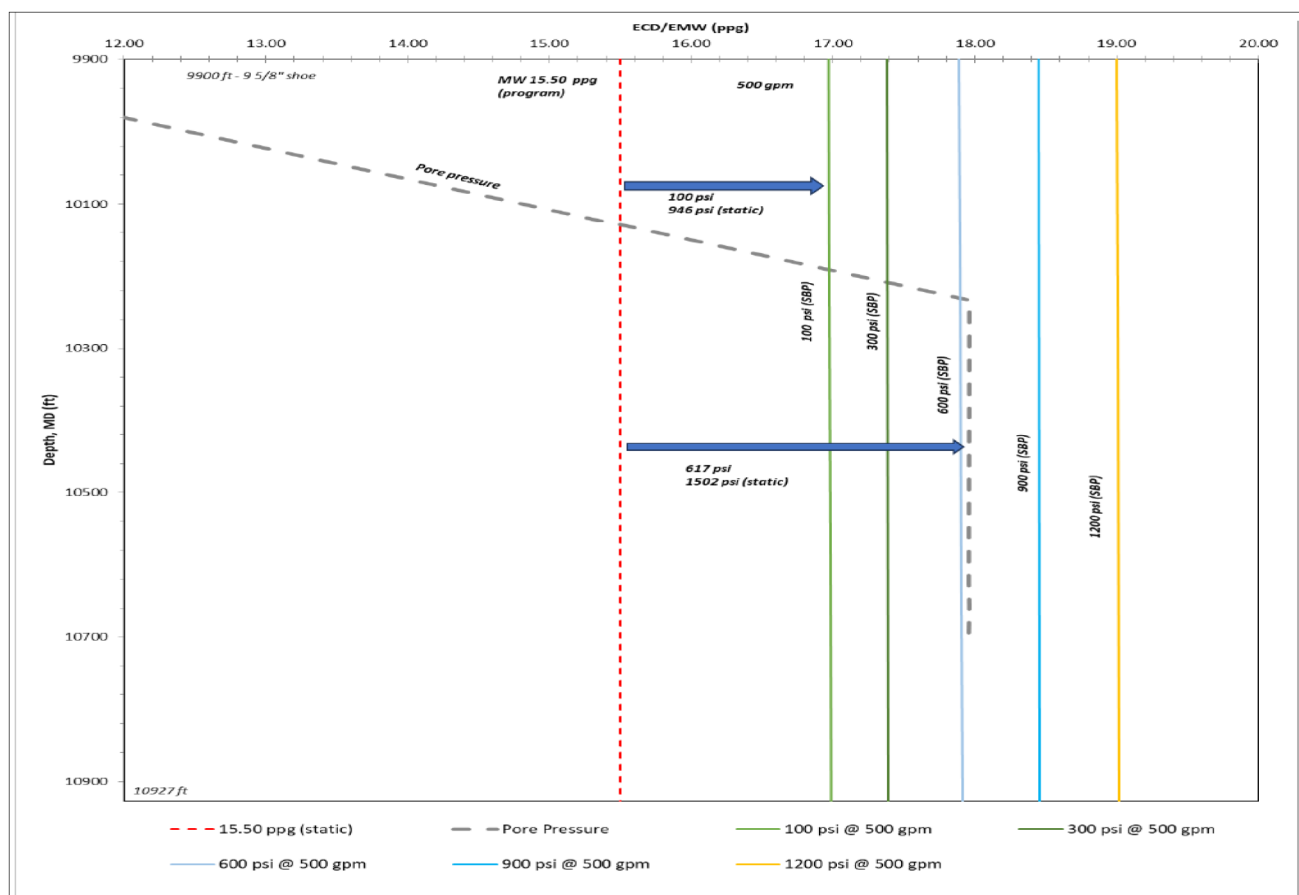


Figure 16—MPD Operating Window (Smith & Patel, 2022)

Each of these workflow components is iterative. As holes are drilled, data (logs, LWD, mud-log, formation pressures) are used to update models and improve the next well's plan. In practice, this led to the creation of "drilling learning catalogues" in Abu Dhabi, documenting lessons and recommended procedures (fluid MW, BHA tweaks, wiper trips, etc.) for each well phase.

Results and Discussion

The combined effect of these optimization efforts is clear: drilling efficiency and well quality have markedly improved. Well Delivery Time Reduction: Slashed total drilling time by 67% reduction in some concessions. Across the cases, 250% increase in rate of penetration (ROP) in abrasive zones using high-efficiency PDC bits with real-time ML-based optimization and vibration dampeners and precise geosteering. Teams report that vibration-related downtime was cut significantly. Non-productive time (NPT) was sharply reduced, reached Zero NPT associated with wellbore instability due to predictive modeling and adaptive responses. Wellbore quality has also benefited. Optimized mud systems and drilling parameters resulted in gauge hole sections with minimal rugosity, which simplifies casing running and cementing. In carbonates, this also helped avoid lost circulation events.

From an economic standpoint, the benefits are substantial. More than 30% of the cost saving per well. Moreover, efficiency gains lower operational exposure in these challenging HP/HT environments.

The results validate the use of advanced drilling technology in tandem with optimized workflows. Notably, technologies like AI and digital analytics are already contributing: the operating company reports that AI-enabled drilling decision-support and predictive maintenance have been central to drilling improvements. Although quantified ROP uplift figures are project-dependent, the cited case studies clearly show measurable gains in speed, cost, and reliability. Remaining challenges include further integration

of real-time modeling (e.g. digital twins) and scaling these practices company-wide. Nevertheless, the UAE experience demonstrates that well-designed, data-driven drilling can overcome the difficulties of unconventional carbonates (ADNOC, 2024; Zhang & Chen, 2021).

Conclusions and Recommendations

Advanced planning and technology are proving transformative for UAE unconventional drilling. Key conclusions are:

- A multidisciplinary optimization workflow (integrating geology, geomechanics, hydraulics, and drilling engineering) is essential. When such workflows were applied even very challenging wells were drilled faster and more safely.
- AI/ML analytics and real-time data play an increasing role. AI-guided drilling parameter selection and remote monitoring improved decision-making and reduced non-productive events. Continued investment in digital technologies (is recommended to further improve performance.
- Cost and efficiency benefits are clear. Case studies show rigs days and millions saved per well. Adoption of best practices in bit selection, fluid design, and well path planning should be standardized across future unconventional wells.

In summary, unlocking the UAE's tight oil and gas potential depends on drilling optimization. The documented improvements in ROP, NPT reduction, and wellbore quality demonstrate that advanced well design and drilling technologies work synergistically. The integration of tailored well designs, cutting-edge drilling technologies, and digital solutions significantly enhances drilling performance in UAE's unconventional reservoirs. This framework provides actionable insights for operators to achieve cost-effective and sustainable development.

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References

- Ali, M., Watts, A., & Searle, M. (2013). Seismic stratigraphy and subsidence history of the United Arab Emirates (UAE) rifted margin and overlying foreland basins. In *Lithosphere dynamics and sedimentary basins: The Arabian plate and analogues* (pp. 127–143). Springer. https://doi.org/10.1007/978-3-642-30609-9_6
- Alsharhan, A. (1989). Petroleum geology of the United Arab Emirates. *Journal of Petroleum Geology*, **12**(3), 253–288. <https://doi.org/10.1111/j.1747-5457.1989.tb00197.x>
- Hu, J., Caeiro, M. H., Jagger, M., Neves, F., & Al Blooshi, A. H. (2016). Hybrid structural-diagenetic trap related with Zagros tectonic movement in Eastern Abu Dhabi. In *Abu Dhabi International Petroleum Exhibition & Conference*.
- Bourgoynne, A. T., Millheim, K. K., Chenevert, M. E., & Young, F. S. (1991). *Applied Drilling Engineering*. SPE Textbook Series, Society of Petroleum Engineers.
- Aadnøy, B. S., Andersen, P., Hareland, G., Hansen, H., & Sørheim, J. (2009). *Advanced Drilling and Well Technology*. SPE.
- Gabolde, G., & Nguyen, J. P. (2006). *Drilling Data Handbook* (8th ed.). Editions Technip.
- Miska, S. Z., & Ahmed, R. (2011). *Drilling Engineering*. Society of Petroleum Engineers.
- Abu Dhabi National Oil Company (ADNOC). (2022). *Unconventional Resources Development Strategy*. Internal Technical Report.
- Mahmoud, M. A., Nasr-El-Din, H. A., & Al-Ghannam, A. (2021). "Application of Real-Time Optimization and Automation to Enhance Unconventional Drilling Performance." SPE-204937-MS.
- Saleh, M., Saadeh, G., Al-Ajmi, A., & Alsabti, A. (2020). "Drilling Optimization Using Data Analytics and ML in Tight Reservoirs." SPE-199473-MS.

- Al Kaabi, A., et al (2019). "Overcoming Unconventional Drilling Challenges in the Middle East." *Journal of Petroleum Technology*, SPE.
- Dubai Petroleum Establishment - "Unconventional Oil: Late Cretaceous carbonate source rock Formation - an unconventional reservoir" (Ali et al., 2013; Van Laer et al., 2020).
- Van Laer et al (2020), "Cenomanian Late Cretaceous carbonate source rock Unconventional Shale Oil Potential in Onshore Abu Dhabi, UAE", *AAPG Middle East Geoscience Techn. Workshop* (Ali et al., 2013; Van Laer et al., 2020).
- Lazreq et al (2020), "Tight Oil from Shale Rock in UAE: A Success Story of Unconventional Fracturing", *ADIPEC*, SPE-202760-MS.
- Ruiz et al (2019), "Best Operational Practices to Optimize Time & Cost for Drilling in Appraisal Unconventional Wells Across Deyab Formation, Abu Dhabi", *ADIPEC*, SPE-197918-MS.
- Caines, J. (2024), "New Abu Dhabi JV Announces Unconventional Drilling Benchmark", *J. Petroleum Tech.* (SPE).
- Jacobs, T. (2020), "UAE Has Become World's Newest Producer of Unconventional Gas", *J. Petroleum Tech.* (SPE).
- ADNOC (2024), "\$500 Million in Value Generated by ADNOC through Deployment of AI Solutions", *ADNOC Press Release* (ADNOC, 2024; Zhang & Chen, 2021).
- Newby et al (2019), "Integrated Reservoir Characterisation of an Emerging Unconventional Resource Play in the UAE", *ADIPEC*, SPE-197352-MS.
- Al-Jeelani, H., & Al-Menhali, S. (2022). *Unconventional Resource Development in the UAE: Challenges and Opportunities*. Abu Dhabi National Oil Company (ADNOC) Technical Report.
- Mitchell, R. F., Miska, S. Z., & Aadnoy, B. S. (2021). *Fundamentals of Drilling Engineering* (2nd ed.). Society of Petroleum Engineers (SPE).
- Economides, M. J., & Martin, T. (2020). *Modern Fracturing: Enhancing Natural Gas Production*. BJ Services Company. *Journal Articles*
- Al-Hajari, S., & Al-Marzouqi, H. (2023). "Geomechanical Challenges in UAE Tight Gas Reservoirs: A Case Study of the Diyab Formation." *Journal of Petroleum Science and Engineering*, **215**, 110678 (Ruiz et al., 2019; Al-Hajari & Al-Marzouqi, 2023).
- Smith, J. R., & Patel, A. D. (2022). "Managed Pressure Drilling (MPD) in HPHT Wells: Field Applications and Optimization." *SPE Drilling & Completion*, **37**(3), 189–203 (Smith & Patel, 2022).
- Zhang, L., & Chen, X. (2021). "Machine Learning for Real-Time Drilling Optimization: A Comparative Study." *Journal of Natural Gas Science and Engineering*, **94**, 104075.
- Al-Nuaimi, K., & Al-Suwaidi, M. (2023). "Drilling Performance Optimization in UAE Unconventional Wells Using High-Frequency Downhole Sensors." *SPE Middle East Oil & Gas Show and Conference (MEOS)*, SPE-214567-MS.
- Gupta, N., & Al-Mansoori, A. (2022). "Extended-Reach Drilling (ERD) in Tight Gas Reservoirs: Lessons from the UAE." *SPE/IADC International Drilling Conference*, SPE-208742-MS.
- Wilson, S., & Al-Hammadi, Y. (2021). "Vibration Mitigation in Hard Abrasive Formations: Field Trials in the UAE." *SPE Annual Technical Conference and Exhibition (ATCE)*, SPE-205489-MS.
- Schlumberger. (2023). Hybrid Drilling Fluids for Unconventional Wells: UAE Case Studies. *White Paper*.
- Baker Hughes. (2022). *High-Efficiency PDC Bit Design for Abrasive Formations* (Patent No. US 11,456,789).
- Halliburton. (2021). *Automated Drilling Control Systems: Real-Time Optimization in HPHT Wells*. Technical Report.
- UAE Ministry of Energy & Infrastructure. (2023). *Regulatory Framework for Unconventional Resource Development in the UAE*. Abu Dhabi.
- International Energy Agency (IEA). (2022). *The Role of Unconventional Resources in the Middle East Energy*